DATA SCIENCE CASE STUDY

Credit card fraud detection

Two days of transactions, two models, and an honest comparison of what actually works.

Rodrigo Ferro · November 2026

THE TASK

Score every transaction by its risk of fraud

283,726
transactions, after removing 1,081 duplicates

473
frauds

0.17%
fraud rate: about 1 in 600

48 h
European cardholders, Sept 2013

Predicting “never fraud” is 99.8% accurate and completely useless. The metric has to focus on the frauds.

EVALUATION DESIGN

Measure it the way it will be used: past → future

Temporal hold-out

Train on hours 0–40. Test once on hours 40–48 (74 frauds).

Forward CV

Every modelling choice: 5 folds, each trained on the past and validated on the next block.

Average precision

How well frauds rank above genuine ones. No-skill baseline: 0.0013, not 0.5.

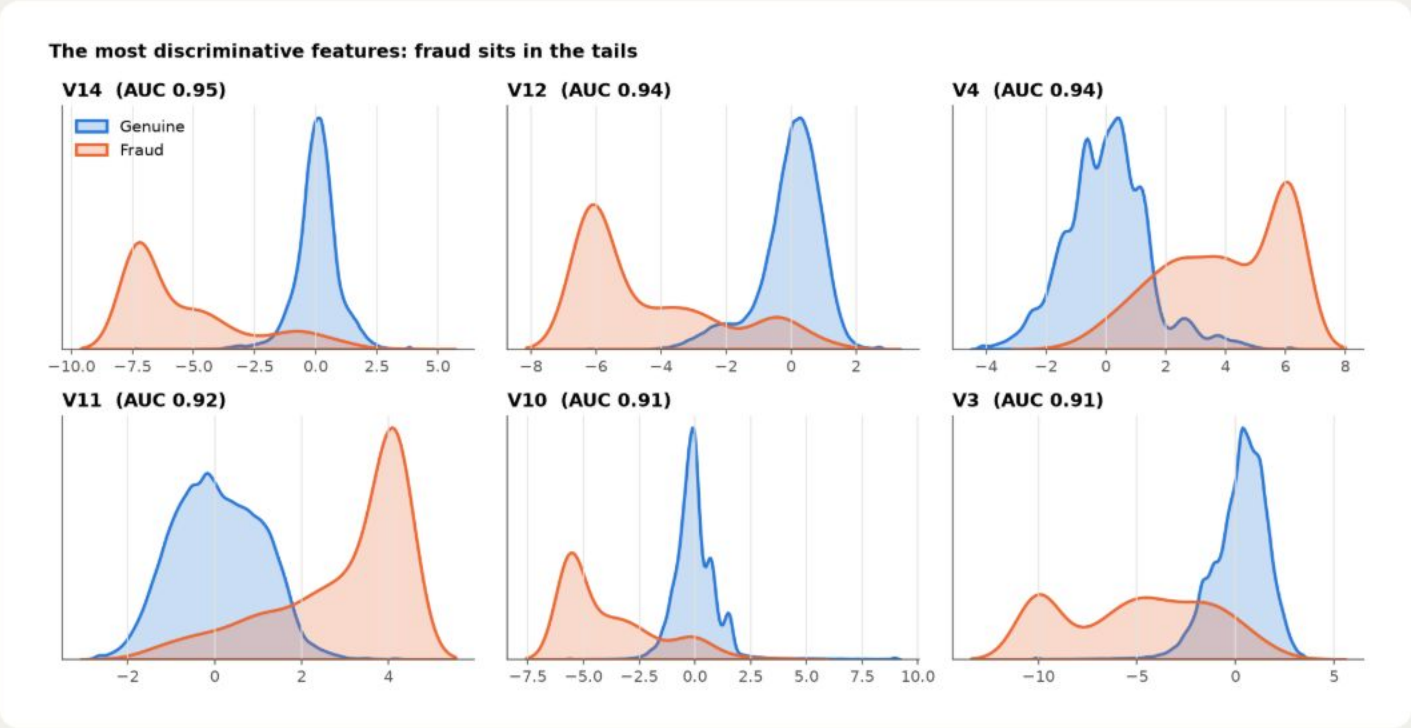
Only 74 test frauds, so every test number comes with a bootstrap 95% confidence interval.

WHAT THE DATA SHOWS

Fraud lives in the tails of a few components

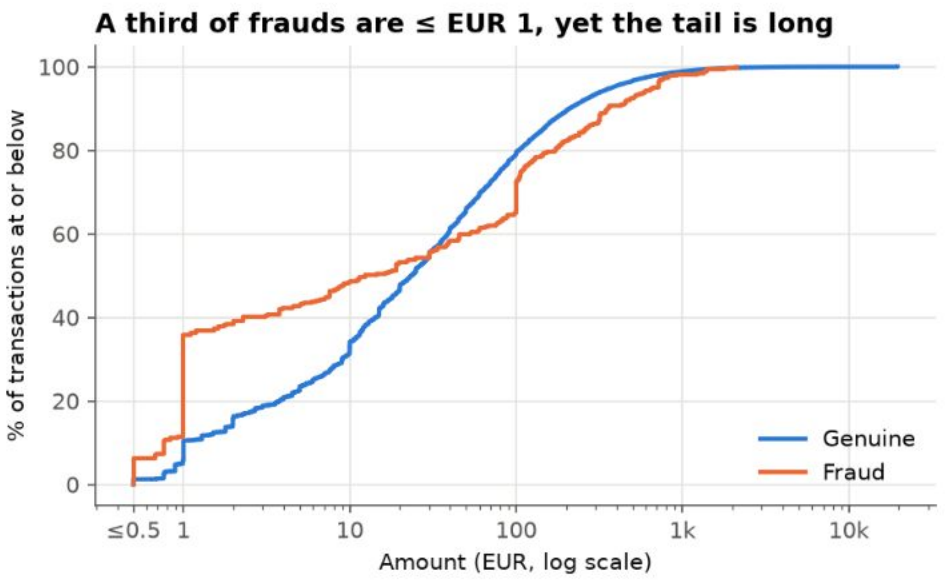
Strong signal. Six components separate the classes on their own, with univariate AUC 0.91–0.95.

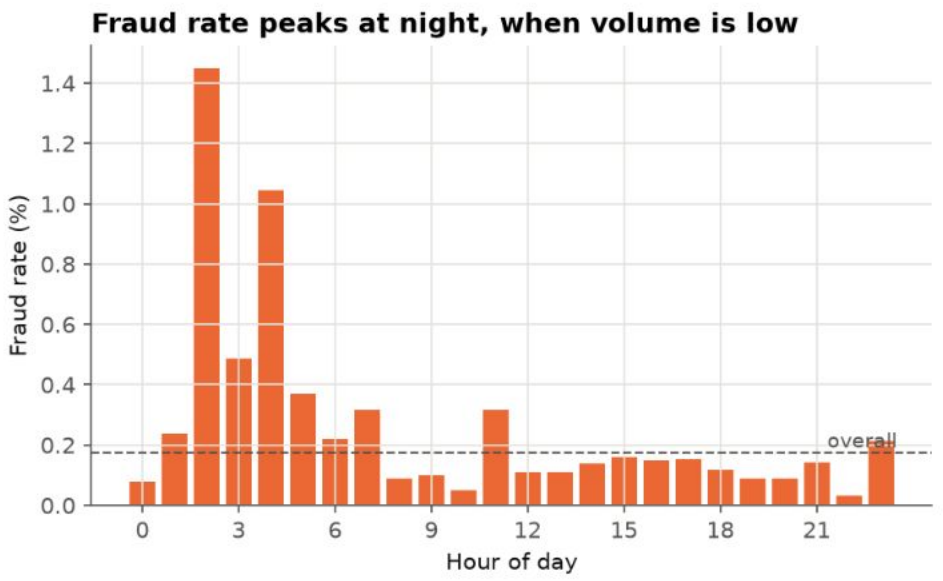
No redundancy. The components are uncorrelated (max |r| = 0.02), so no feature selection is needed.



WHAT THE DATA SHOWS

Small amounts and night hours stand out





36% of frauds are ≤ €1 (vs. 11% of genuine), consistent with card testing.

Night (0–6h) carries 10% of the volume but 29% of the frauds.

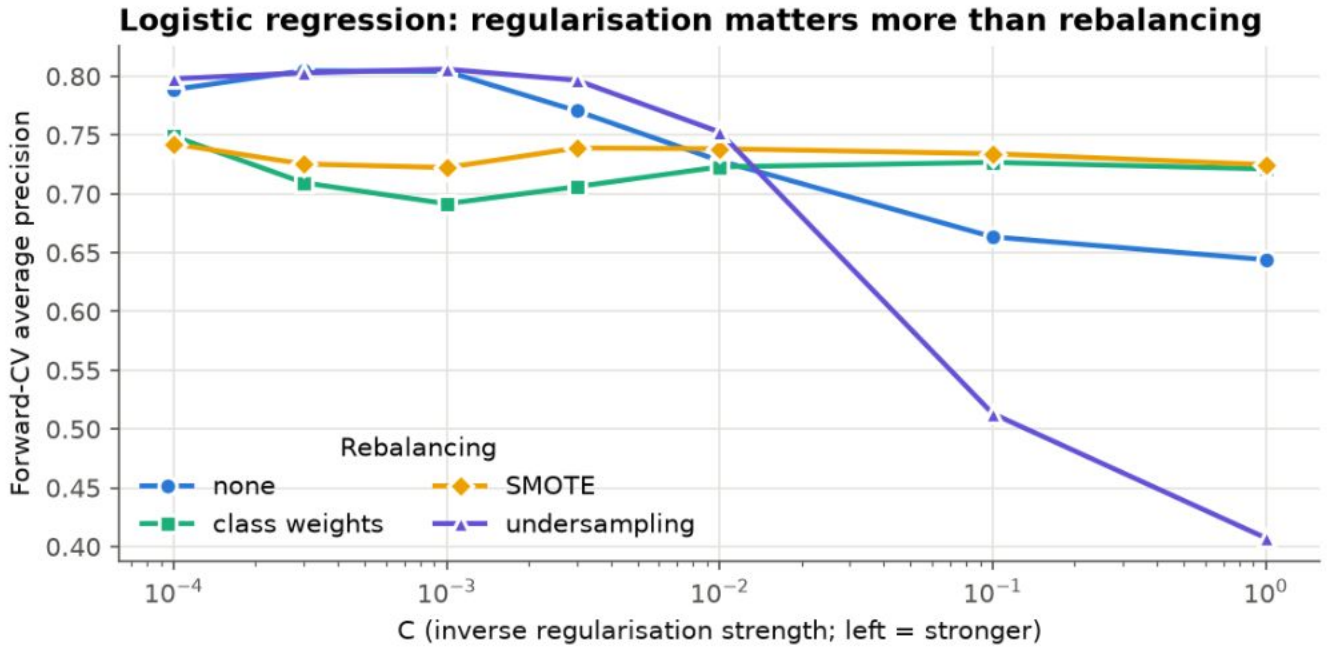
MODELLING · LOGISTIC REGRESSION

Regularisation mattered more than rebalancing

0.64 → 0.81

Forward-CV AP, from the default C=1 to strong L2 ($C=3\cdot10^{-4}$).

SMOTE, class weights and an anomaly score added nothing once the model was regularised. Undersampling was erratic.



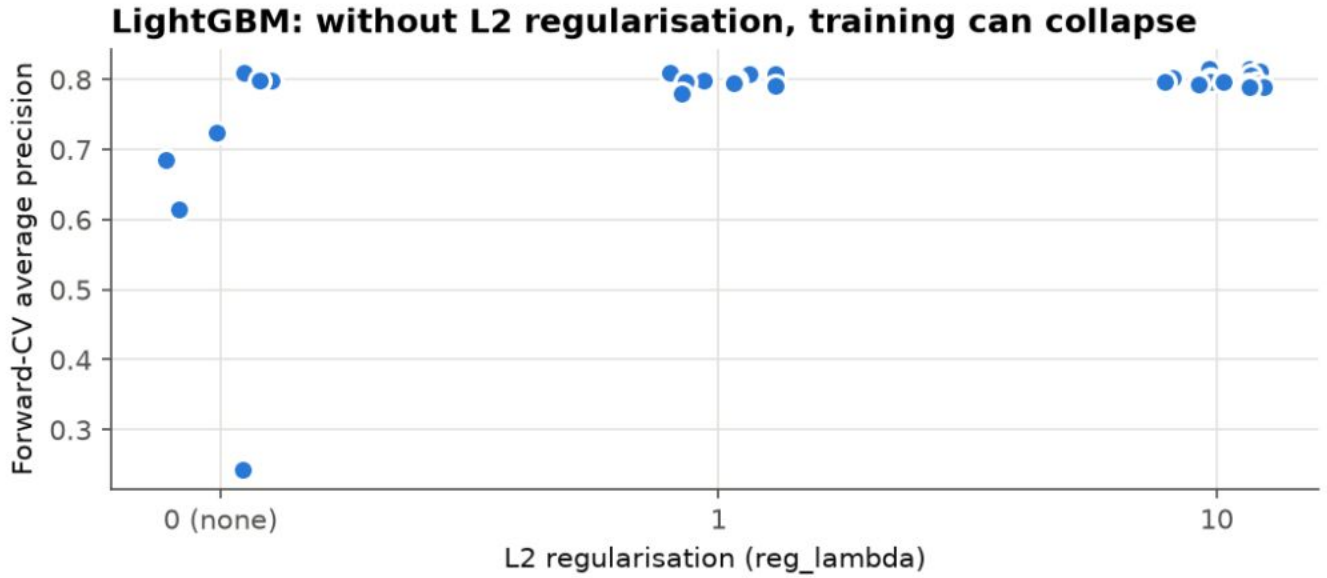
MODELLING · LIGHTGBM

The same lesson for gradient boosting

0.78–0.82

Forward-CV AP of all 23 configurations with L2 regularisation.

Without it, training can collapse: as low as 0.24.



TEST-PERIOD RESULTS

The two models are tied

Logistic regression

0.808

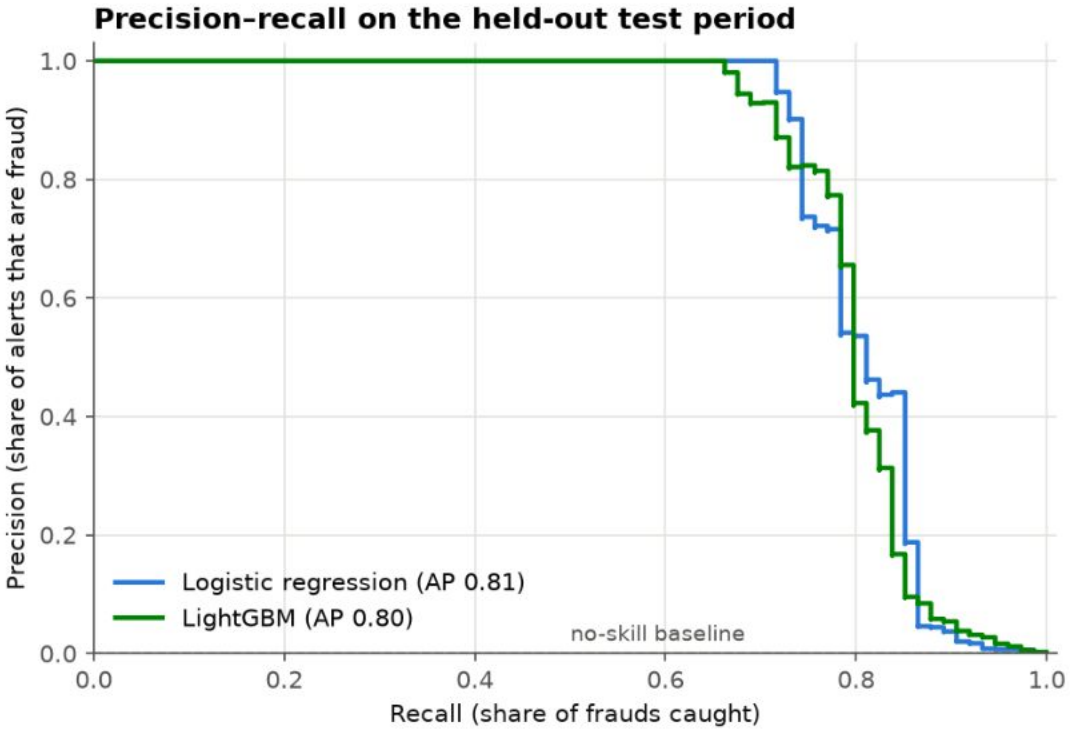
test AP · 95% CI 0.73–0.89

LightGBM

0.801

test AP · 95% CI 0.72–0.89

Paired difference: −0.008 (−0.030 to +0.013). Forward CV called it the same way, so the simpler model is preferred.



THE KEY METHODOLOGICAL LESSON

Shuffled cross-validation would have told the wrong story

Average precision	Shuffled CV	Forward CV	Test
Logistic regression, C=1	0.742	0.644	0.753
Logistic regression, selected	0.744	0.805	0.808
LightGBM	0.863	0.815	0.801

Shuffled CV overstated LightGBM by ~0.06 and couldn’t see the regularisation effect. Forward CV matched the test for every model.

FROM SCORES TO ALERTS

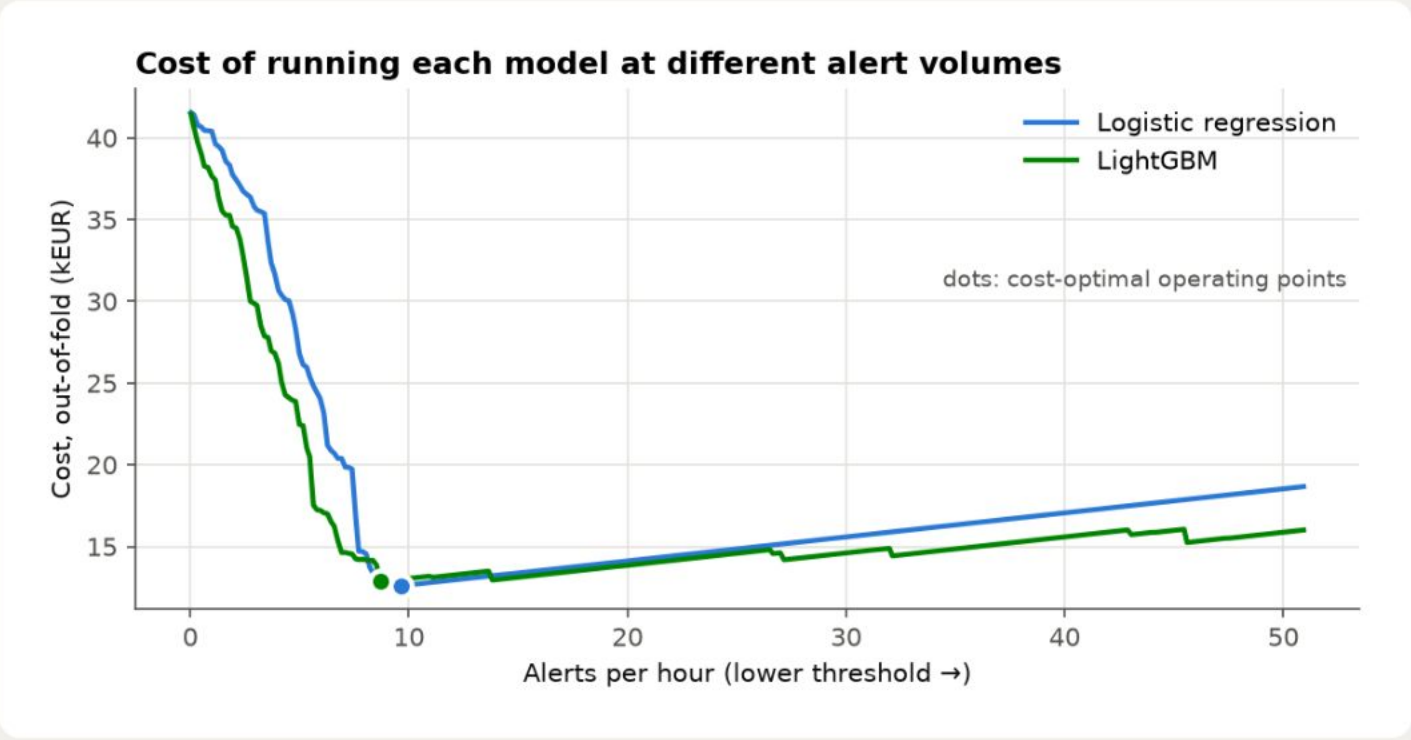
Pick the threshold with an explicit cost

~10 / hour

Cost-optimal alert volume for both models.

Cost = €5 per review [assumed] + the amount of every missed fraud.

Chosen on training data only. Stable from €1 to €50 per review.



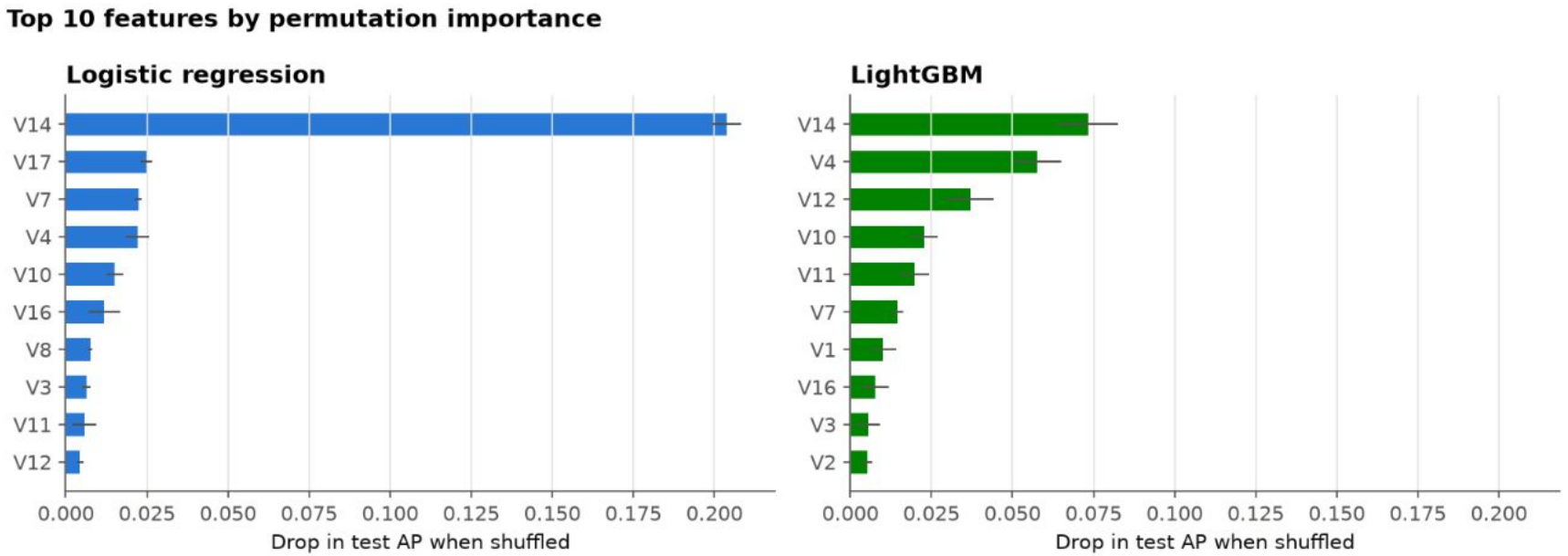
TEST PERIOD · ≈ 8 HOURS · 74 FRAUDS

Either model cuts the cost of fraud by 60–65%

At the threshold chosen on training data	Logistic regression	LightGBM
Alerts	79 (~10 / hour)	64 (~8 / hour)
Frauds caught	57 of 74	54 of 74
False alarms	22	10
Precision (95% CI)	0.72 (0.62–0.82)	0.84 (0.75–0.93)
Recall (95% CI)	0.77 (0.67–0.87)	0.73 (0.63–0.83)
Fraud value caught	€5,327 (69%)	€5,049 (65%)
Total cost (no model: €7,728)	€2,796	€2,999

WHAT DRIVES THE PREDICTIONS

Same features, different concentration



Logistic regression leans on V14: shuffling it alone costs 0.20 AP.

LightGBM spreads its reliance, so it is less exposed to drift.

RECOMMENDATION

Ship the simple model, keep the challenger running

1

Deploy logistic regression

Same performance, cheaper to serve, monitor and explain. About 10 alerts per hour.

2

LightGBM in shadow mode

Promote it if its cleaner alert stream holds up on live data.

3

Monitor weekly

Alert volume, precision and score drift. Re-tune the threshold as real costs arrive.

LIMITATIONS AND NEXT STEPS

Promising, but tested on two days of data

Limitations

- Only 74 test frauds: wide intervals, untested drift.
- Simple cost model: no chargebacks, customer friction or capacity limit.
- Anonymised features limit interpretation.

Next steps

- Walk-forward evaluation over months.
- Real review costs and an alert-capacity constraint.
- A simple ensemble of the two models.
- Card-level behaviour features from raw data.